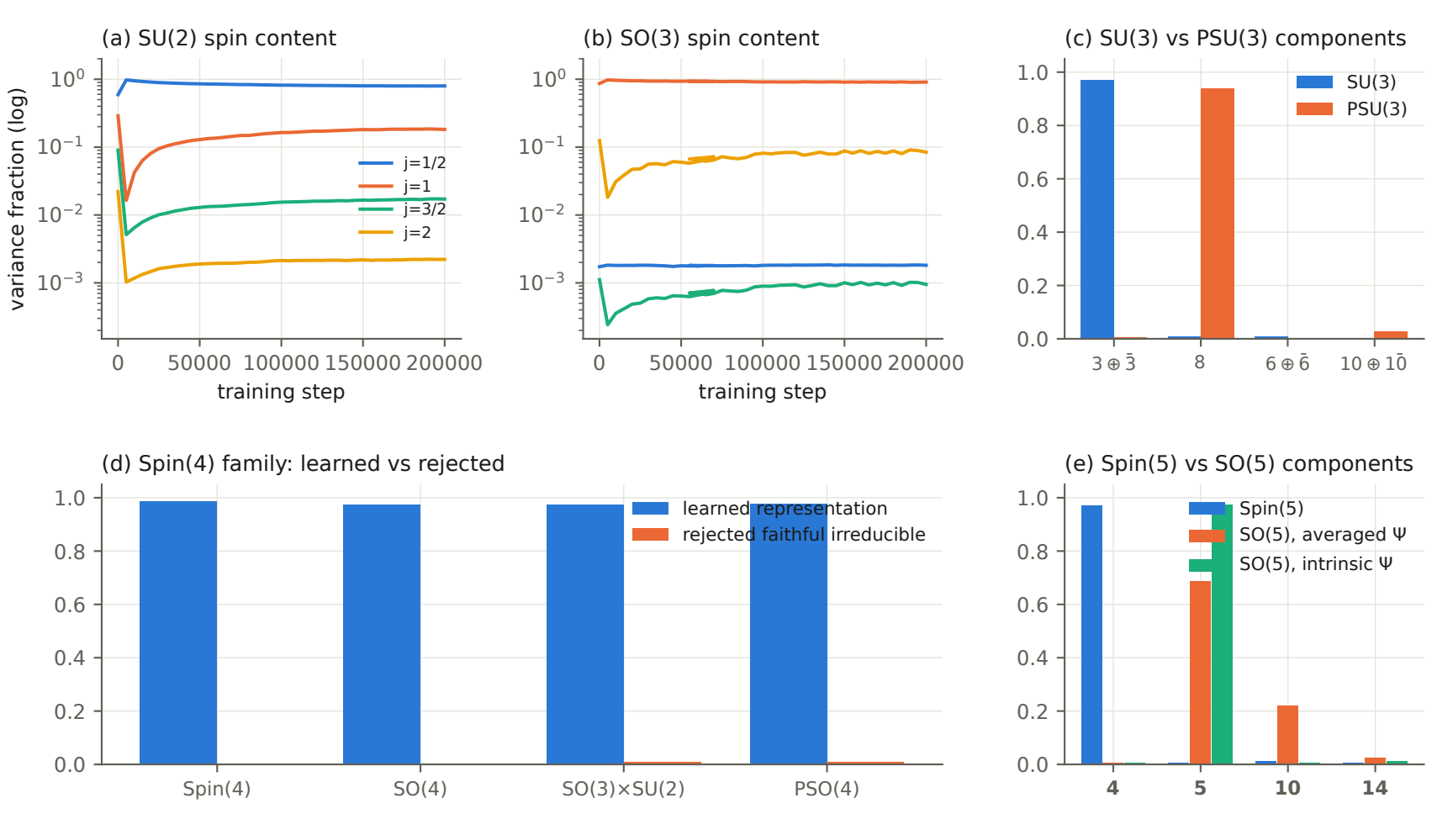
(a) SU(2) spin content
variance fraction (log)
10^0
10^{-1}
10^{-2}
10^{-3}
j=1/2
j=1
j=3/2
j=2
0
50000
100000
150000
200000
training step
(b) SO(3) spin content
10^0
10^{-1}
10^{-2}
10^{-3}
0
50000
100000
150000
200000
training step
(c) SU(3) vs PSU(3) components
1.0
0.8
0.6
0.4
0.2
0.0
SU(3)
PSU(3)
$3 \oplus \bar{3}$
8
$6 \oplus \bar{6}$
$10 \oplus \bar{10}$
(d) Spin(4) family: learned vs rejected
1.0
0.8
0.6
0.4
0.2
0.0
learned representation
rejected faithful irreducible
Spin(4)
SO(4)
SO(3)×SU(2)
PSO(4)
(e) Spin(5) vs SO(5) components
1.0
0.8
0.6
0.4
0.2
0.0
Spin(5)
SO(5), averaged Ψ
SO(5), intrinsic Ψ
4
5
10
14